

Temperature-dependent Sellmeier equation for the refractive index of stoichiometric lithium tantalate

Ariel Bruner, David Eger, Moshe B. Oron, Pinhas Blau, and Moti Katz

Electro-Optics Division, Soreq Nuclear Research Center, Yavne 81800, Israel

Shlomo Ruschin

Department of Electrical Engineering—Physical Electronics, Tel Aviv University, Tel Aviv 69978, Israel

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We present a temperature-dependent Sellmeier equation for the refractive index of stoichiometric LiTaO₃. The extraordinary refractive index, for the range 0.39–4.1 μm and for temperatures of 30–200 °C, are based on previously published data [Jpn. J. Appl. Phys. **41**, 465 (2002)] and on measured data derived from quasi-phase-matched (QPM) resonances. We used the new Sellmeier coefficients that we obtained to calculate the QPM wavelengths for an optical parametric oscillator (OPO) based on periodically poled stoichiometric LiTaO₃ pumped at 1064 nm. The measured wavelengths of the OPO were in good agreement with our predictions. © 2003 Optical Society of America

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Periodically poled LiTaO₃ is a most attractive optical nonlinear material because of its wide transparency range (0.28–5.5 μm), highly nonlinear coefficients,¹ and high resistance to optical damage.² Commercially available LiTaO₃ is mostly of the congruent type (CLT). Recently, stoichiometric LiTaO₃ (SLT) crystals were grown.^{2,3} The coercive field in SLT (1.7–4.5 kV/mm) is ~ 1 order of magnitude lower than in CLT.² This low value facilitates the electrical poling of thicker crystals for quasi-phase-matched (QPM) frequency conversion.³ The low coercive field of the stoichiometric crystals is apparently related to their lower defect density. Additionally, these crystals show lower power threshold and higher conversion efficiency in QPM optical parametric oscillation (OPO),³ a higher photorefractive damage threshold, and lower green-induced infrared absorption.²

Knowledge of the refractive-index dispersion (Sellmeier equation) of nonlinear crystals is essential for the design of frequency-conversion devices. For these applications the refractive index of the crystal has to be known within an accuracy of 0.0001.

The refractive indices of SLT were previously measured only at room temperature and in the range 0.440–1.050 μm .⁴ The Sellmeier equation that was obtained was not suitable for predicting the phase-matching wavelengths at different temperatures and beyond 1.050 μm . Recently we published some preliminary results of refractive-index dispersion.⁵

In this study we derived a temperature-dependent Sellmeier equation for SLT crystals. The equation is applicable in the range 0.39–4.1 μm and at temperatures of 30–200 °C. The dispersion equation is deduced from high orders of QPM second-harmonic generation (SHG) and QPM OPO measurements and from absolute refractive-index values that were previously measured at room temperature.⁴ The Sellmeier equation that was derived facilitates the design of efficient QPM frequency-conversion devices for several applications.

For the electrical poling process we used commercial (Deltronic Crystals Company and the Oxide Company) Z-cut stoichiometric LiTaO₃ plates, 18 mm long and 1 mm thick. The crystals were grown by the double-crucible Czochralski method,⁶ and their Curie temperature was ~ 690 °C. Photoresist stripes were produced on the c^+ face parallel to the x crystal axis, by standard photolithography techniques and then covered with a thin titanium layer. Photolithography masks with various periods of 26.5 to 29.5 μm were used with a duty factor of 50%. A plane uniform electrode was deposited upon the c^- face. We achieved electrical poling by applying high-voltage pulses to the polar crystal surfaces. The total switching charge required for complete poling was 65 $\mu\text{C}/\text{cm}^2$, and the coercive field was ~ 4 kV/mm. Etching the samples in HF revealed a uniform periodic structure through the entire crystal depth and over the whole interaction length.

To determine the refractive indices we measured QPM second harmonic (SH) and OPO wavelengths. The crystals were placed in a temperature-stabilized oven with 0.1 °C accuracy. First we measured several orders of SHG while we scanned the laser wavelength, using a tunable Ti:sapphire laser source with a tuning range of 0.76–1.04 μm . The laser beam was focused by an $f = 175$ mm lens into the crystal. The beam waist radius at the crystal center was 60 μm . We also used a narrow-linewidth, tunable OPO system (MOPO-HF, Spectra-Physics Lasers) pumped by a tripled single-longitudinal mode Nd:YAG source (Quanta Ray Pro 250, Spectra-Physics Lasers) in the 0.85–1.85 μm range. The measurements were made at temperatures of 30–200 °C at intervals of 10 °C with a Burleigh WA-2000 wavelength meter with a spectral accuracy of ~ 1 pm.

Scanning of the laser beam parallel to the x axis of the crystal and perpendicular to the crystal surface showed that the SH power did not vary in all areas, indicating that the domain reversal is homogeneous across the entire thickness. With 0.5 mW of

power at $0.808\ \mu\text{m}$ coupled into the crystal we obtained $60\ \mu\text{W}$ of blue light in ninth-order phase matching for SHG. This is equivalent to $3.85\%/W$ conversion efficiency in first-order SHG.

An optical parametric oscillator was constructed with periodically poled (PP) SLT crystals. It was pumped by a Q -switched Nd:YAG laser (Spectra-Physics; pulse width, 24–25 ns; repetition rate, 1 kHz; maximum pulse energy, 1.5 mJ). The beam was focused by a 100-mm focal-length lens into a focal spot of $100\text{-}\mu\text{m}$ waist radius ($1/e^2$) in the cavity center. The cavity length was 47 mm and consisted of two plane mirrors. The input mirror was highly reflecting at the signal and at the idler wavelengths. The output mirror was highly reflecting at the pump and the signal and highly transmitting at the idler wavelength. The idler wavelength was measured with a calibrated Jobin-Yvon-Spex 500M monochromator.

We used the experimental measurements of the pump, SH, signal, and idler wavelengths for a series of period-temperature pairs to determine refractive indices in the $0.39\text{--}4.1\text{-}\mu\text{m}$ range. First we attempted to represent the Sellmeier equation in a form similar to that which was used for CLT,⁷ namely,

$$n_e^2(\lambda, T) = A + \frac{B + b(T)}{\lambda^2 - [C + c(T)]^2} + \frac{E}{\lambda^2 - F^2} + D\lambda^2, \quad (1)$$

where Sellmeier coefficients A , B , C , D , E , and F are constants, c and b are temperature dependent, T is the temperature in degrees centigrade, and λ is the wavelength in micrometers.

The QPM condition for SHG is defined by the following equation:

$$\Lambda - \frac{m\lambda_m}{2[n(\lambda_m/2) - n(\lambda_m)]} = 0, \quad (2)$$

where Λ is the period length, m is the SH order, λ_m is the fundamental wavelength for m th-order phase matching, and $n(\lambda_m/2)$ and $n(\lambda_m)$ are the refractive indices for the frequency doubled and the fundamental wavelengths, respectively.

For an OPO process the QPM OPO condition is as follows:

$$\frac{1}{\Lambda} - \left[\frac{n(\lambda_{\text{pump}})}{\lambda_{\text{pump}}} - \frac{n(\lambda_{\text{signal}})}{\lambda_{\text{signal}}} - \frac{n(\lambda_{\text{idler}})}{\lambda_{\text{idler}}} \right] = 0. \quad (3)$$

We derived the Sellmeier coefficients from the measurements by substituting the Sellmeier notation, $n(\lambda, T)$ into Eq. (2) for SHG or into Eq. (3) for OPO. We adjusted the parameters by least-squares data fitting, using the Gauss-Newton method to provide the phase matching.

The phase-matching measurements are sensitive mostly to the dependence of wavelength on refractive indices (the dispersion). We obtained an absolute Sellmeier equation by including absolute refractive-index data at room temperature⁴ in the data fitting.

To take into account the crystal's thermal expansion, we used the following expression⁸:

$$\Lambda(T) = \Lambda(25^\circ\text{C})[\alpha(T - 25^\circ\text{C}) + \beta(T - 25^\circ\text{C})^2], \quad (4)$$

where α and β are the thermal expansion coefficients and T is the crystal temperature. As we did not find any information about SLT thermal expansion, we used the coefficients given for CLT crystals, $\alpha = 1.6 \times 10^{-5}$ and $\beta = 7 \times 10^{-9}$. We used this equation assuming that there is not a significant difference between the two crystals.

The Sellmeier equation can predict the QPM wavelength for a given period length with an accuracy of $\pm 2.5\ \text{nm}$ in the visible and of $\pm 20\ \text{nm}$ in the IR. Then we tried to improve the fitting. Adding another pole to the Sellmeier equation reduced the sum of the quadratic errors by a factor of 7 and enhanced the wavelength accuracy. However, letting the second and third poles be temperature dependent did not statistically improve the fit results. The final form of the Sellmeier equation that we obtained is

$$n_e^2(\lambda, T) = A + \frac{B + b(T)}{\lambda^2 - [C + c(T)]^2} + \frac{E}{\lambda^2 - F^2} + \frac{G}{\lambda^2 - H^2} + D\lambda^2. \quad (5)$$

We performed a least-squares data fit on all the data simultaneously and obtained the Sellmeier coefficients for SLT that are given and compared in Table 1 with those of CLT.⁵ This Sellmeier equation provides the QPM wavelength within an accuracy of $\pm 1.5\ \text{nm}$ for SHG and within $\pm 8\ \text{nm}$ for OPO near $4\ \mu\text{m}$.

The additional pole, at $H = 7.99\ \mu\text{m}$, represents an optical transition at this wavelength of a relatively low absorption strength, G . A similar pole found for LiNbO₃ was attributed to multiphonon absorption.⁹ The SLT refractive indices are lower than those of the CLT crystal, as is reasonable if one takes into account that SLT material has fewer defects than CLT. It should also be noted that measurements

Table 1. Sellmeier Coefficients for SLT and CLT^a

Coefficient	SLT	CLT
A	4.502483	4.514261
B	0.007294	0.011901
C	0.185087	0.110744
D	-0.02357	-0.02323
E	0.073423	0.076144
F	0.199595	0.195596
G	0.001	
H	7.99724	
$b(T)$	3.483933×10^{-8} $(T + 273.15)^2$	1.82194×10^{-8} $(T + 273.15)^2$
$c(T)$	1.607839×10^{-8} $(T + 273.15)^2$	1.5662×10^{-8} $(T + 273.15)^2$

^aSource: Ref. 5.

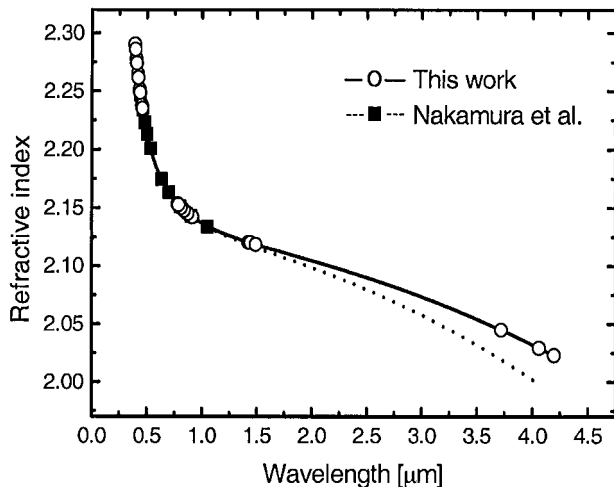


Fig. 1. Sellmeier equation obtained for SLT (solid curve) at room temperature. Dashed curve, the SLT dispersion curve given in Ref. 4.

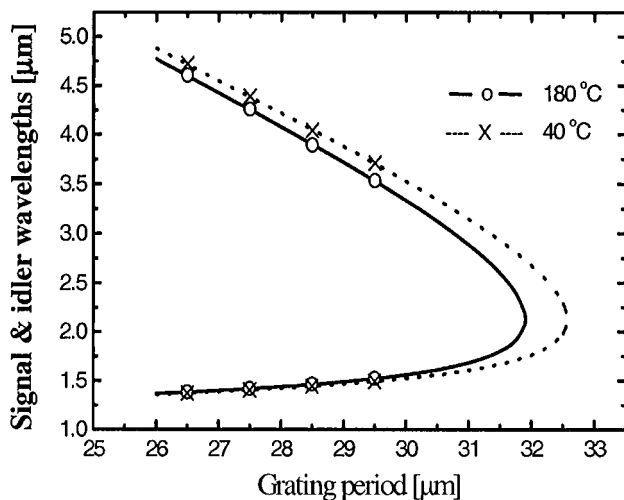


Fig. 2. Measured data (circles and crosses) and calculated tuning curves for a 1.064- μm -pumped PP SLT optical parametric oscillator at temperatures of 40 and 180 °C, that produced the Sellmeier coefficients.

could vary slightly for crystals of different quality or from different vendors.

As shown in Fig. 1, at wavelengths shorter than 1.05 μm the SLT dispersion curve obtained in the research reported in this Letter coincides with the curve given in Ref. 4 (the dashed curve). But, in the IR range, a difference between the curves appears that increases with wavelength. This difference results from the fact that the equation given in Ref. 4 is suitable only for the 0.44–1.05- μm range.

Figure 2 shows the measured data and calculated tuning curves for a 1.064- μm -pumped optical parametric oscillator at temperatures of 40 and 180 °C, for the fit results of Table 1. From the Sellmeier

coefficients we calculated the QPM wavelengths of the idler and the signal for a PP SLT optical parametric oscillator pumped at 1064 nm and checked them experimentally. We used the OPO described above setup with a 22-mm-long crystal and a 28.08- μm period. The crystal was antireflection coated at the pump and signal wavelengths. The reflection at the idler wavelength was $\sim 10\%$ for each facet. The idler's tuning range was 4.1828–3.9898 μm for temperatures of 40–200 °C. The wavelengths obtained were in good agreement with our predictions.

At a repetition rate of 1 kHz we obtained an oscillation threshold of $\sim 10 \text{ MW/cm}^2$ and maximal idler efficiency of 17%, which corresponds to a total internal conversion efficiency of more than 60%. We measured 0.5 W of average output power near 4- μm at a repetition rate of 10 kHz. The idler's efficiency scarcely changed at various temperatures above 70 °C, and no photorefractive damage was observed. These results demonstrate that PP SLT crystals are suitable for high-power applications at temperatures lower than those that can be tolerated by CLT or LiNbO₃.

In conclusion, we have obtained a temperature-dependent dispersion curve for SLT in a 0.39–4.1- μm range. The resultant Sellmeier equation can accurately predict the QPM wavelengths. As PP SLT may be important for high-power applications, we believe that this equation can be helpful for the designer of nonlinear optical devices based on this material. As SLT has the advantage of transparency to 270 nm, we are conducting further research to expand the Sellmeier equation into the UV range.

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